3-6-21

Circular Motion

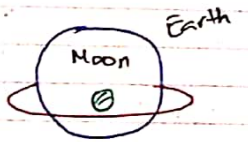
Circular Motion

Motion in a circular path, semicircular path or along a curved path.

* If complete circular path
* If semicircular path as it's considered as a part of circle
* If curve can be considered as a part of circle then circular motion is done.

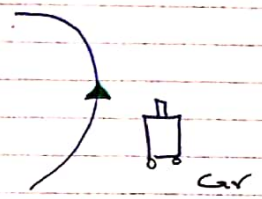
Example

1-



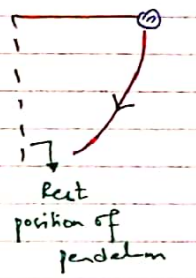
* Earth, moon, sun along with other planets etc are considered as "perfect sphere"

2-



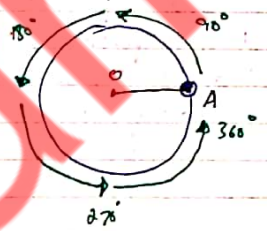
* orbit are considered as a "perfect circle"

3-



* Circular motion is also called "Angular Motion"
  → During circular motion angle is produced with respect to center of circle

e.g.



* Angle can be more than 360°

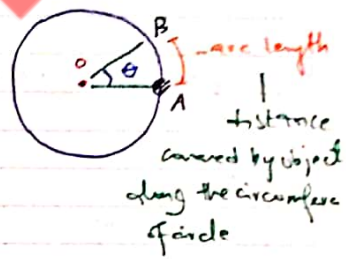
linear motion and angular displacement for angular motion.

Angular Displacement (θ)

Angle produced during circular motion is called "Angular Displacement"

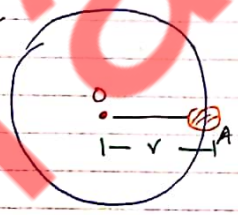
→ Represented by "θ"

→ SI unit = rad (ration)



1 rad = 57.3°

Question



Prove

$\theta = 2\pi$ rad

$\Rightarrow 2 \times 3.142 \times 57.3$

$\Rightarrow 360.00°$

proven

$$\boxed{S = r\theta}$$

* S = Arc length
* r = radius of circle
* θ = Angular displacement

Ration

Angular displacement is 1 ration if arc length is equal to the radius of the circle

$\theta = \frac{S}{r}$

$\theta = \frac{2\pi r}{r}$

$\boxed{\theta = 2\pi \text{ rad}}$ — proven

$360 = 2\pi$ rad

$\frac{360}{2 \times 3.142}$ = rad

$\boxed{57.3 = 1 \text{ rad}}$ — proven

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→ $2\pi rad = 360^\circ$

* Always put numerical value of π whenever used.

Angular Velocity

Rate of change of Angular displacement

OR

Angle produced in unit time is called "Angular Velocity".

$$\text{Angular velocity} = \frac{\text{Angular displacement}}{\text{time}}$$

$$\boxed{\omega = \frac{\theta}{t}}$$

Omega

unit = $\frac{rad}{s}$ or $\frac{rev}{min}$ — if given, convert into rad/s

↑ SI unit

→ $1 rev = 2\pi rad$

e.g) $\omega = 33 \frac{rev}{min}$

$\frac{33 \times 2 \times 3.142 \times \cancel{\ }}{60} \frac{rad}{s}$

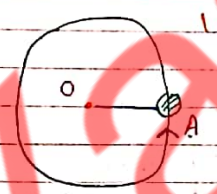
$\Rightarrow 3.45 \frac{rad}{s}$

Time Period (T)

Time taken by an object to complete one circular motion is called "Time Period"

Frequency of Circular Motion ($f = \frac{1}{T}$)

No of circular motion produced in one second.



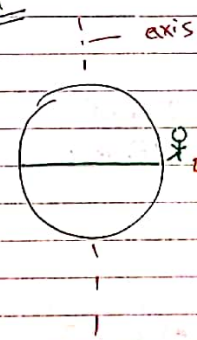
1 complete circular motion

$\omega = \frac{\theta}{t}$

$$\boxed{\omega = \frac{2\pi}{T}}$$ → Relation of T & Angular velocity

↓ valid for all types of circular motion

Question



Angular velocity ⇒ ω

$\omega = \frac{\theta}{t}$

$= \frac{2\pi}{24 \times 3600}$

$= 7.2 \times 10^{-5} rad s^{-1}$

Note

Time taken by Earth for 1 complete rotation = Time taken by moon to complete it's circular motion

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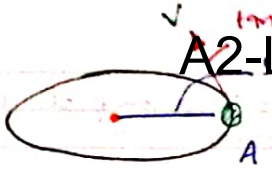


(linear velocity)

Before release of string $\Rightarrow d = 2\pi r$ (circumference)

Linear velocity = $\frac{\text{Distance}}{\text{time}}$

$V = \frac{2\pi r}{T}$

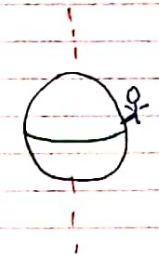
$V = \left(\frac{2\pi}{T}\right) r$

$V = \omega r \Rightarrow \boxed{V = r\omega}$ linear velocity

$\boxed{V = \frac{2\pi}{T}}$ Angular velocity

Q.1

$r = 6.4 \times 10^6 \, m$

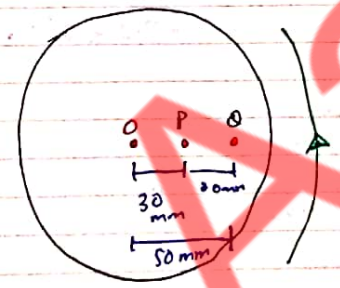


$V = r\omega$

$= 6.4 \times 10^6 \times 7.26 \times 10^{-5}$

$= 464.6 \approx 465 \, m/s$

Q.2



33 rev/min

a) $\omega = ?$

b) $\Delta\omega = ?$

c) $\Delta V = ?$

b $\Rightarrow$ i $\Rightarrow$ $\Delta\omega$ = zero

* P & Q are points of the same disc the same (ω)

ii $\Rightarrow$ $V = r\omega$

$= \frac{50}{1000} \times 3.45$

$= 0.1725 \, ms^{-1}$

$V = r\omega$

$= \frac{30}{1000} \times 3.45$

$= 0.1035 \, ms^{-1}$

$\Rightarrow 0.1725 - 0.1035$

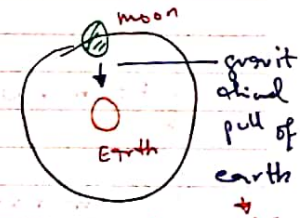
$= 0.069 \, ms^{-1}$

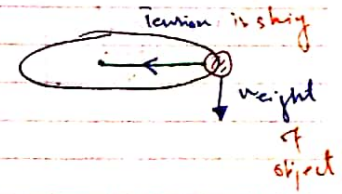
10-6-21 **Circular Motion** Thursday

Centripetal Force

1. It is an unbalanced resultant force
2. It is directed towards centre of circle

* Whenever circular motion occurs, at every instant the direction of object changes therefore velocity changes at every instant. Due to change in velocity, acceleration is produced, this is called "centripetal acceleration" that is directed towards the centre of circle.





$\boxed{a = V\omega}$

$\boxed{a = r\omega^2}$

$\boxed{a = \frac{V^2}{r}}$

* V = linear velocity

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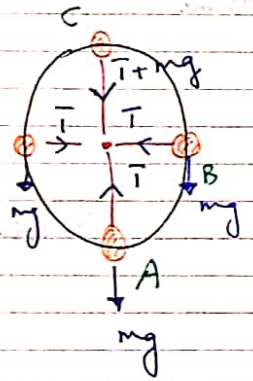
→ To find centripetal force we have to find all the forces acting on the object in the given situation

→ Then the force that contains the properties of centripetal force will be the centripetal force

→ Moon has a long distance from planets such as sun etc thus their pull force is said to be negligible.

→ In different situations the source of centripetal force can be different.



At point A = $\boxed{T - mg}$ — provides C.F

At point B = T — provides C.F

At point C = $\boxed{T + mg}$ — provides C.F

At point D = T — provides C.F

* $\boxed{F = mv\omega}$

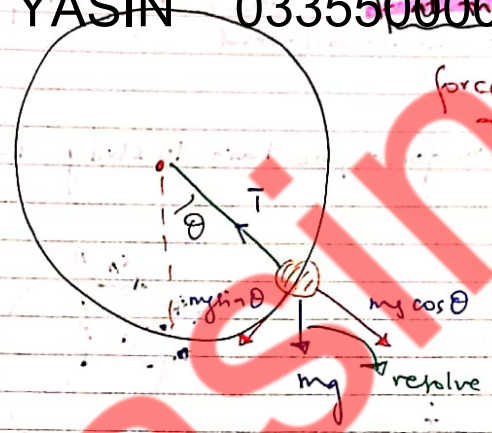
$F = ma$

$\boxed{F = mr\omega^2}$

To find the magnitude of centripetal force.

$\boxed{F = \frac{mv^2}{r}}$

Resolve the forces and then apply the general rule



$T - mg\cos\theta = \frac{mv^2}{r}$

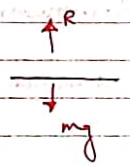
$\boxed{T - mg\cos\theta = C.F}$

$\boxed{T = \frac{mv^2}{r} + mg\cos\theta}$

general expression

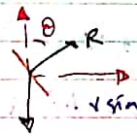
* lift is always perpendicular to the wings of the plane

a) If at constant speed/height



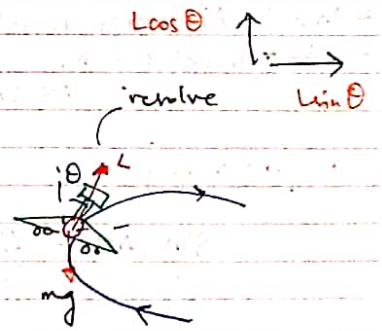
$R = mg$ for passenger

b) If doing a curve path



for passenger

$v\sin\theta$ — centripetal force



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* For the same [illegible] the source of centripetal force for 2 objects can be different

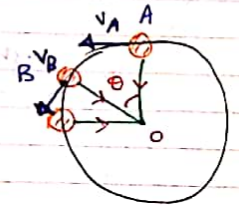
** Before centripetal force, we have to study this ↓

Acceleration

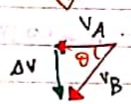
↓

Change in velocity

Speed | Direction of Motion



⇓



$$\boxed{V_A + \Delta V = V_B}$$

or

$$\boxed{\Delta V = V_B - V_A}$$

Centripetal Acceleration

Acceleration of object performing circular motion

→ It is always directed towards centre of the circle

* Even if speed or direction of motion changes, acceleration is produced in the change in velocity.

* In ΔV ↓ As change in velocity is directed downwards so is the acceleration.

∴ ΔV direction ✓ Acceleration direction

→ $\boxed{a = v\omega}$

$\sin(\Delta\theta) = \frac{\Delta V}{V_B}$ $\qquad \Delta\theta = \frac{\Delta V}{V}$

$\sin(\Delta\theta) = \frac{\Delta V}{V}$ $\qquad \Delta V = V \cdot (\Delta\theta)$

$\sin(\Delta\theta) \simeq \Delta\theta$ $\qquad a = \Delta V / \Delta t$

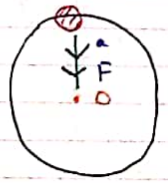
$\frac{\Delta\theta}{\Delta t} \Rightarrow \omega$ $\qquad \rightarrow \omega = \frac{V}{r}$ (from $V = r\omega$)

$\therefore \boxed{a = V\omega}$

→ $V = r\omega \Rightarrow a = V\omega$

$a = (r\omega) \times \omega$

$\boxed{a = r\omega^2}$



→ $\omega = \frac{V}{r} \Rightarrow a = V\omega$

$a = V\left(\frac{V}{r}\right)$

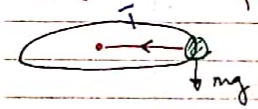
$\boxed{a = \frac{V^2}{r}}$

14-6-21 **Circular Motion** Monday

Motion in Horizontal Circle

Case 1

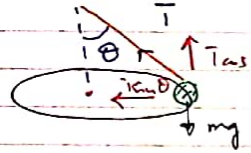
• tension is in string source of centripetal force



→ String is also horizontal

Case 2

→ $T\sin\theta$ is providing centripetal force to the object



→ string is inclined

* Both cases, the height is constant

$T = C.F$

$$\boxed{T = \frac{mv^2}{r}}$$

* m - mass of object
v^2: linear velocity
r = Radius of circle

$T\sin\theta = C.F$

$$\boxed{T\sin\theta = \frac{mv^2}{r}}$$

* Motion if height is constant then "Horizontal Circular Motion"

* " " " " not constant then "Vertical Circular Motion"

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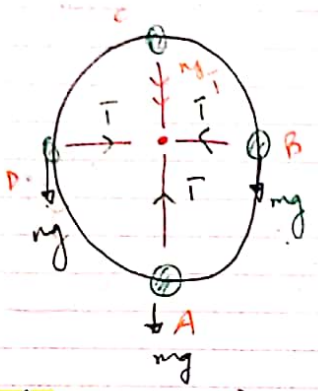
Motion in Vertical circle



Point A (Lowest point)

$T - mg = C.F$

$T - mg = \frac{mv^2}{r}$

$$\boxed{T_{max} = \frac{mv^2}{r} + mg} \quad (1)$$

Point B

$T = C.F$

$$\boxed{T = \frac{mv^2}{r}} \quad (2)$$

Point C (Highest Point)

$T + mg = C.F$

$T + mg = \frac{mv^2}{r}$

$$\boxed{T_{min} = \frac{mv^2}{r} - mg} \quad (3)$$

Point D

$T = C.F$

$$\boxed{T = \frac{mv^2}{r}} \quad (4)$$

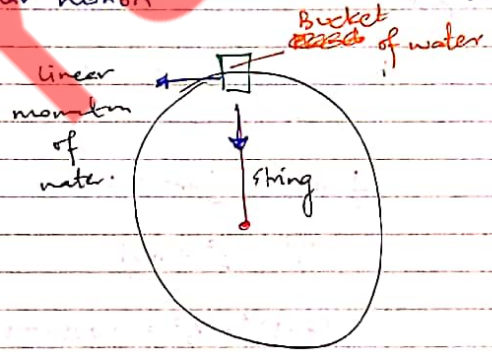
* A to C - Max to min
  C to A - Min to Max

* Use expressions accordingly.

* Moon is trying to move to perpendicular direction simultaneously thus the resultant force will be along the curved path

* For circular motion, we need centripetal force and linear momentum. These both need to compromise to each other.

* Due to linear momentum / velocity moon follows circular motion



Track of roller coaster

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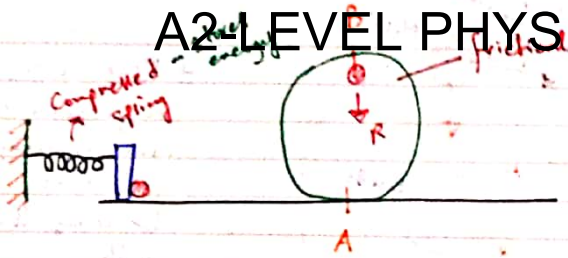
Q.1



Energy = 0.09 J
mass = 0.025 kg
radius = 0.1 m

a) $v_1 = ?$ — upon leaving the spring

b) $v_2 = ?$ — At point B

c) $R = ?$ — Force exerted at ball at B

Vertical circle

* Can be solved by "Law of Conservation of T.E" (K.E = G.P.E) and equation of "centripetal force"

a) $K.E = \frac{1}{2}mv^2$

$0.09 = \frac{1}{2} \times 0.025 v^2$

$0.18 = 0.025 v^2$

$v^2 = \sqrt{7.2}$

$v = \pm 2.68 \, ms^{-1}$

b) T.E at A = T.E at B

$\frac{1}{2}mv_A^2 = (K.E + G.P.E)$

c) $R + mg = C.F$

$R = C.F - mg$ → v_A is used as at B

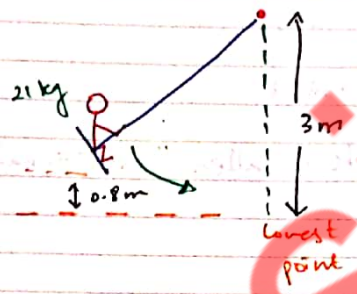
$R = \frac{mv_A^2}{r} - mg$

$R = \frac{0.025 \times \quad}{0.1} - 0.025 \times 9.81$

$R =$

Q.2

Lowest Point



a) $v = ?$

b) $a = ?$

c) $R = ?$

d) Linear momentum will be more the h which it'll travel in a straight path and leave the swing.

a) Gain in K.E = Loss in P.E

$\frac{1}{2}mv^2 = mgh$

$\frac{1}{2}mv^2 = gh$

$v^2 = 2gh$

$v = \sqrt{2gh}$

$v = \sqrt{2 \times 9.81 \times 0.8}$

$v = 3.96 \, ms^{-1}$

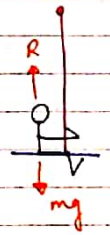
b) $\frac{v^2}{r} = a$

$\frac{3.96^2}{3} = a$

$a = 5.22 \, ms^{-2}$

* a remains constant as v & r are same

c)



$R - mg = C.F$

$R = C.F + mg$

$= \frac{mv^2}{r} + mg$

$= \frac{21 \times 3.96^2}{3} + 9.81 \times 21$

$= 109.7 + 206.01$

$= 315.7 = 315 \, N$

* 2 forces acting on child at lowest point
a) N. Contact
b) Weight of boy

* No tension, as tension acts on objects directly attached to the string (seat in this case). child isn't directly attached

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